NATIONAL UNIVERSITY OF SINGAPORE
SCHOOL OF COMPUTING
MID-TERM TEST FOR
CS3244: MACHINE LEARNING

4 October 2018 **Time Allowed: 60 minutes**

INSTRUCTIONS TO CANDIDATES

1. This test paper contains TWENTY ONE (21) multiple choice questions, TWO (2) identity questions and comprises SEVEN (7) printed pages, including this page, worth a total of 60 marks (~1 minute/mark).
2. This is a closed book test. You are not to use search or any precomputed, programmed routines if you are using a digital device.
3. Some MCQs are worth more than others. This is dependent on the section.
4. There are several versions of this test. Your neighbours have a different version of the test. Cheating by looking at others' responses will not help you.
5. The form allows for you to shade in multiple responses and a *nil* response where indicated. If a question doesn't state that multiple responses applies, then mark exactly one choice. E.g.:
   - If multiple responses are sanctioned by the question, and you feel Responses A, B, C apply to a question, shade all three of A, B and C:
     ● ● ● Ⓓ Ⓔ
   - If a *nil* response is sanctioned by the question, and you feel no responses apply, omit any markings:
     Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
6. You should use a pencil to take the exam. Pen marks may not record properly.
7. You must return this question paper with your MCQ form.
8. You may use the backs of the pages as scratch paper, as they will be disregarded, *unless* specifically noted by you.
9. You are asked to write your name and matriculation number below for recordkeeping purposes. Only the MCQ form will be graded, unless otherwise noted by you.
10. If you have any written explanation that you feel merits the grading staff's attention, please highlight it in the appropriate question at the end of the assessment.

Name: _____

Matriculation Number: | | | | | | | | | | (please print legibly)

I certify that the answers I have provided in this midterm to be my original work and that I did not receive nor attempt to obtain answers from any other source. By signing, I certify that I followed the assessment procedures written above. I understand that anyone caught cheating will receive a 0 mark for the assessment and may face disciplinary action.

Signature: _____ **Date**: _____

Q1: Exam Identity (Part 1)

Please shade this MCQ as Ⓐ ● Ⓒ Ⓓ Ⓔ. This is an ungraded item. This is mandatory to ensure that your midterm is graded against the appropriate key.

Section 1. Multiple Choice and Multiple Response Questions.
Each question is worth 2 marks; 15 questions. Total: 30 marks.

Q2-5: Learning Paradigms

These responses pertain to the questions below:

A. Recognizing whether fast food is present in a picture
B. Transcribing the dialog between two specific people from the acoustic signal from microphones in a noisy MRT platform.
C. Planning module lecture times to minimize conflicts for students.
D. Answering multiple choice questions from midterms
E. Guessing which NUS courses a student takes in a current semester, based on their past transcript (modules and grades).

Mark all of the responses that could be construed as instances of the ML paradigm below. If none fit, do not mark any.

Q2. Supervised Classification.
Q3. Supervised Logistic Regression (Binary or Multinomial).
Q4. Unsupervised Machine Learning.
Q5. Reinforcement Learning.

Q6. The notion of a cost function is applicable to which tribe of machine learning? Mark one response.

A. Bayesians.
B. Connectionists.
C. Symbolists.
D. Evolutionaries.
E. Applicable to more than one tribe.

Q7. What is the minimum and maximum number of 2-D training points in 1 nearest neighbor that could be responsible for the following binary classification borders with 2 input dimensions? Assume that training examples have distinct input values. Mark one response.

A. 10, ∞
B. 9, ∞
C. 10, 10
D. 8, 10
E. 9, 9

Q8. Which of the following is **true** of *Noisy Targets*? Mark all responses that apply. If none fit, do not mark any.

A. We use the deterministic target function $f(x)$.
B. Noise = $y - f(x)$.
C. $P(y|x)$ is used instead of $f(x)$.
D. Noisy target = $\mathbb{E}[y|x] + (y - f(x))$.
E. None of the above.

Q9. Which of the following is **true** for *cost functions*? Mark all responses that apply. If none fit, do not mark any.

A. It measures the amount of error made by our hypothesis.
B. Which cost function to use depends on the problem.
C. Plausible measures are easy to use but cannot be proved.
D. Convenient measures have analytic proof.
E. Squared error is an example of a plausible measure.

Q10. Which of the following applies to PLA in the general case? Mark all responses that apply. If none fit, do not mark any.

A. A plausible stopping criterion is that we terminate after reaching a threshold cost value.
B. A plausible stopping criterion is that we terminate after a fixed number of iterations (large enough to converge).
C. In each iteration, we update the weight based only on misclassified example(s).
D. It will never reach a state where all examples classified correctly, if the dataset is not linearly separable.
E. After updating the weights, the accuracy of PLA will stay the same or improve.

Q11. How many marks a student can get in a competitive exam, based on her/his hours of study can be solved using a _____ model. Mark one response.

A. Linear Regression.
B. Logistic Regression.
C. Naïve Bayes.
D. None of the above.
E. A, B and C.

Q12. In logistic regression, we often use the negative of the log-likelihood of the training data as the cost function. During training, when we get a lower negative of the log-likelihood, it indicates which of the following? Mark one response.

A. The logistic regression model now fits the training data worse.
B. The logistic regression model now fits the training data better.
C. We will get a better classification accuracy in the training data.
D. We will get a better classification accuracy in the testing data.
E. The cost function should maximize the negative of the log-likelihood, not minimize it.

Q13. Suppose we are using the batch gradient descent (GD) to optimize the logistic regression model. We observe that it takes approximately T seconds to run each iteration of the GD. If we change the optimization method to stochastic gradient descent (SDG), how long it will take (approximately) to run one iteration then? Recall we use the variables m and n to denote the number of training instances, and the number of features per instance. Mark one response.

A. $\frac{T}{n}$

B. $\frac{T}{m}$

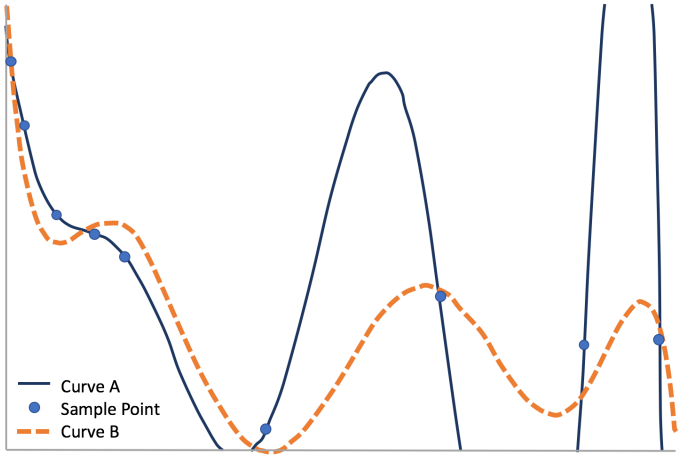
C. $\frac{T}{\log n}$

D. $\frac{T}{log\ m}$

E. None of the above.

Q14. From the following graph, identify which is target function and which is the approximation function. Note that the data points are noisy. Mark all responses that apply.

A. Target function: Curve A.
B. Approximation function: Curve B.
C. Target function: Curve B.
D. Approximation function: Curve A.
E. Impossible to tell.



Q15. To overcome *underfitting*, which of the following is/are helpful? Mark all responses that apply.

A. Having more data points.
B. Increasing the complexity of hypothesis.
C. Decreasing complexity of the target function.
D. Adding noise.
E. None of them.

Q16. If k is increased as in k-fold cross validation, the learned hypothesis h will tend more towards which of the following phenomenon? Mark one response.

A. Overfitting.
B. Underfitting.
C. Neither overfitting nor underfitting.

Q17: Exam Identity (Part 2)

Please shade this MCQ as Ⓐ Ⓑ ● Ⓓ Ⓔ. This is an ungraded item. This is mandatory to ensure that your midterm is graded against the appropriate key.

Section 2. Computation Questions. Marks differ per question. Total: 30 marks.

Q18-21: (16 marks) Linear Regression with Regularization
In this problem we are working on linear regression with regularization on points in a 2-D space. Figure 2 plots linear regression results on the basis of three data points, (0,1), (1,1) and (2,2), with different regularization penalties.

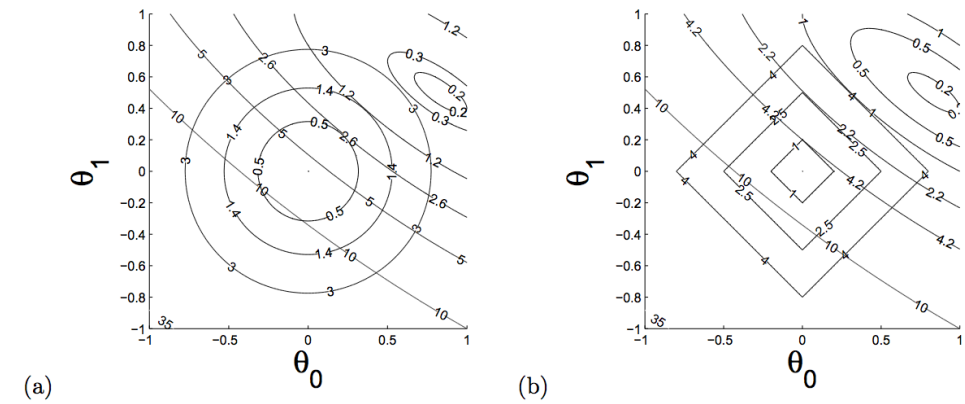


Figure 1. Contour plots for the linear regression problem with (a) L^2 or (b) L^1 regularization where the ellipsis correspond to the squared error term, and circles/diamond correspond to the regularization penalty term.

Assign each plot in Figure 2 to one (and only one) of the following regularization methods identified by Q18-21 (a one-to-one mapping). You may find Figure 1 helpful.

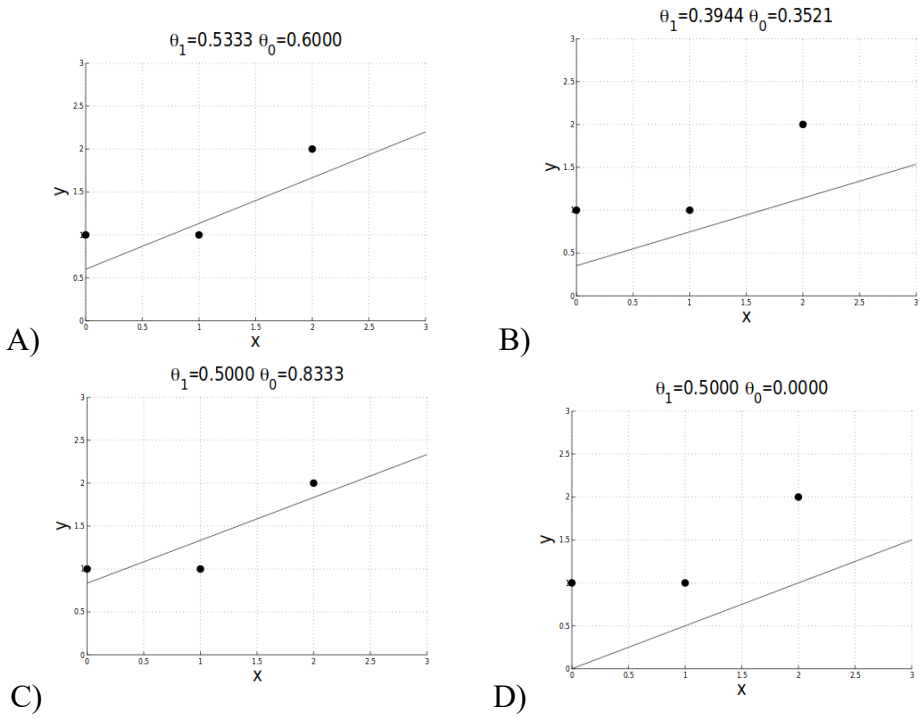


Figure 2. Linear regression with and with regularization.

Q18. (3 marks) No regularization. *I.e.,* $\sum_{j=1}^{3}\left(y_j - x_j\theta_1 - \theta_0\right)^2$

Q19. (5 marks) L^2 reg. with λ = 5. *I.e.,* $\sum_{j=1}^{3}\left(y_j - x_j\theta_1 - \theta_0\right)^2 + \lambda(\theta_1^2 + \theta_0^2)$, where $\lambda = 5$.

Q20. (3 marks) L^1 reg. with λ = 5. *I.e.,* $\sum_{j=1}^{3}\left(y_j - x_j\theta_1 - \theta_0\right)^2 + \lambda(|\theta_1| + |\theta_0|)$, where $\lambda = 5$.

Q21. (5 marks) L^2 reg. with λ =1. *I.e.,* $\sum_{j=1}^{3}\left(y_j - x_j\theta_1 - \theta_0\right)^2 + \lambda(\theta_1^2 + \theta_0^2)$, where $\lambda = 1$.

Q22-Q23: Naïve Bayes (14 marks)

The following questions refer to this data table of instances with categorical inputs and output, which tabulates the temperature on the week before the midterm (binary outcome: Hot or Cold), whether a rare Pokémon was sighted earlier at the Arts canteen (binary: Yes or No) and whether the CS 3244 staff went to go to the Arts Canteen for their lunch (binary: Yes or No).

Temperature	Rare Pokémon Sighted at Arts?	Eat at Arts?
Hot	Yes	Yes
Cold	Yes	Yes
Cold	No	Yes
Hot	No	No
Cold	No	Yes

Recall that Naïve Bayes determines $P(y|X) = \prod_{j=1}^{n} P(x_j|y)P(y)$. That is, it makes an independence assumption between the features of X, and calculates the likelihood as a term.

Q22: (6 marks) Today it's hot and rare Pokémon have been not been sighted at Arts. Is it more likely that the CS3244 staff go to eat at Arts? Mark one choice.

A. Yes. More likely to go to Arts.
B. No. More likely to go somewhere else.
C. Unsure. There's an equal probability.
D. Impossible to know, given the data.

Q23: (8 marks) Today it's hot and some students saw the CS3244 staff go to the Arts Canteen for lunch. Calculate the probability that rare Pokémon have been sighted in Arts. Mark one choice.

A. Between 0.0 to 0.4.
B. Between 0.4 to 0.6.
C. Between 0.6 to 0.8.
D. Between 0.8 to 1.0.
E. Not possible as there's no causal relation. Where CS 3244 staff go does not cause Pokémon to show up.

End of Midterm Assessment. Do not write beyond this area.